

JiHo Lee

Ph.D. Student in Computer Science Department @ Virginia Tech
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INTERESTS

Network Security and Privacy, VR/XR Systems, Traffic Analysis, Behavioral Privacy, Edge Computing, Wireless Ad-Hoc Networks

EDUCATION

Virginia Tech,

- Ph.D. Student in Computer Science

Aug 2024 – Present

University of Virginia,

- M.S in Computer Science

Aug 2021 – May 2024

Ewha Womans University, Seoul, South Korea

- M.S. in Computer Science and Engineering
• Academic Course Highlights: Edge Computing, Algorithms, Embedded Systems, Data Mining

Mar 2019 – Feb 2021

Dankook University, Yongin, South Korea

- B.S. in Software Science
• Completed Accreditation Board for Engineering Education of Korea

Mar 2014 – Feb 2019

RESEARCH EXPERIENCE

Synergistic Networking and Augmented Intelligence Lab (SNAIL), Virginia Tech Aug 2024- Present

PhD Student (Advisor: *Prof. Bo Ji*)

- Research Project: Network-layer Privacy in Virtual Reality
 - Led research on passive network-layer privacy risks in multi-user VR applications, focusing on motion telemetry leakage from network traffic, and published a paper (*USENIX Security'26*).
 - Designed PACMO, a passive, encoding-agnostic attack that reconstructs VR head and hand motion from plaintext UDP payloads without application access or message-format knowledge.
 - Measured traffic from 50 VR applications and found that motion-bearing multi-user uplink traffic commonly uses high-frequency plaintext UDP.
 - Showed that reconstructed motion enables privacy attacks, including 96.2% user identification and 69.4% virtual typing inference accuracy.
 - Focus: Security and Privacy, Network Traffic Analysis, Virtual Reality
- Research Project: Multi-Modal Models on Edge
 - Participated in research on efficient multi-modal models for distributed multi-task inference on the edge devices, and published paper (*IEEE ICDCS'25*)

Research Assistant, University of Virginia

Sep 2021 – May 2024

- Research Project: Automated Database Generation for Web Application Security Testing
 - Participated in research on synthesizing databases to facilitate security testing of PHP web applications, and published paper (*NDSS'23*)
 - Implemented a concolic execution engine to explore diverse program paths and generate corresponding database constraints.
 - Conducted security analyses with 17 real-world PHP applications, demonstrating improved code and query coverage compared to existing state-of-the-art techniques, and discovering 33 previously unknown vulnerabilities.
- Research Project: Anti-Forensic Server-side Attack via Fail-free Dynamic State Machine
 - Participated in designing Dazzle-attack, a server-side attack technique via specialized state-machine against existing malware detection and obfuscation techniques (*WISA'22*)

Intelligent Networked Systems Laboratory, Ewha Womans University

M.S. Student (Advisor: *Prof. HyungJune Lee*)

Mar 2019 – Feb 2021

- Research Project: On-Device AI with Federated Deep Learning

- Led research on on-device distributed deep learning over wirelessly connected edge devices, and published paper (*IEEE Access*'22)
- Proposed semi-distributed deep learning architecture based on model parallelism, which is resilient against link volatility or device failure environments
- Achieved high inference quality on visual classification tasks with resilience to dynamic changes at resource-constrained edge devices via simulation-based experiments
- Focus: Federated Machine Learning, Wireless Networks, Edge Computing
- Research Project: Infrastructure-Less Vehicle Traffic Density Estimation
 - Extended previous research work on traffic density estimation using distributed vehicles in VANET within time-deadline, and published paper (*IEEE TVT*'20)
 - Designed traffic density estimation system using collaborative distributed multiple vehicle samplers based on past collected information and future possibility
 - Improved estimation performance compared to counterpart algorithms, with a factor of 9.5
 - Focus: Vehicular Ad-Hoc Networks, Traffic Density Estimation

Research Intern (Advisor: *Prof. HyungJune Lee*)

Sept 2018 – Feb 2019

- Research Project: Infrastructure-Less Vehicle Traffic Density Estimation
 - Participated in research on traffic density estimation through lightweight V2V packet probing without relying on infrastructure under time-deadline
 - Focus: Vehicular Ad-Hoc Networks, Traffic Density Estimation

PUBLICATIONS

CONFERENCE

- [1] **JiHo Lee**, JinYi Yoon, Tijay Chung, Brendan David-John, and Bo Ji, "Motion in the Clear: Reconstructing VR User Behavior from Network Traffic," *Proceedings of the USENIX Security Symposium*, 2026.
- [2] JinYi Yoon, **JiHo Lee**, Ting He, Nakjung Choi, and Bo Ji, "S2M3: Split-and-Share Multi-Modal Models for Distributed Multi-Task Inference on the Edge," *Proceedings of IEEE International Conference on Distributed Computing Systems (IEEE ICDCS)*, 2025.
- [3] An Chen, **JiHo Lee**, Basanta Chaulagain, Yonghwi Kwon, and Kyu Hyung Lee, "SynthDB: Synthesizing Database via Program Analysis for Security Testing of Web Applications," *The Network and Distributed System Security Symposium (NDSS)*, 2023.
- [4] Bora Lee, Kyungchan Lim, **JiHo Lee**, Chijung Jung, Doowon Kim, Kyu Hyung Lee, Haehyun Cho, and Yonghwi Kwon, "Dazzle-attack: Anti-Forensic Server-side Attack via Fail-free Dynamic State Machine," *World Conference on Information Security Applications (WISA)*, 2022.

JOURNAL

- [1] **JiHo Lee**, JeiHee Cho, and HyungJune Lee, "StitchNet: Distributed On-Device Model Partitioning over Edge Devices under Volatile Wireless Links," *IEEE Access*, 2022.
- [2] JinYi Yoon, **JiHo Lee**, Nayoung Han, and HyungJune Lee, "A Survey on Parallel Deep Learning," *The Journal of the Korean Institute of Communications and Information Sciences (KICS)*, Vol. 46, No. 10, pp. 1604-1617, 2021 (KCI Journal).
- [3] Christina Suyong Shin, **JiHo Lee**, and HyungJune Lee, "Infrastructure-Less Vehicle Traffic Density Estimation via Distributed Packet Probing in V2V Network," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 10, Oct 2020.

HONORS AND AWARDS

- CCI SWVA Cyber Innovation Scholars Program, Commonwealth Cyber Initiative
(Amount of award: \$1,500) Apr 2026 – Mar 2027
- CCI SWVA Cyber Innovation Scholars Program, Commonwealth Cyber Initiative
(Amount of award: \$1,500) Apr 2025 – Mar 2026
- Research Assistant, University of Virginia Aug 2021 – May 2023
- Research Assistant, Ewha Womans University 2019
- Scholarship for Academic Excellence, Dankook University Spring, Fall 2017
For outstanding academic performance in the department of software science

- National Science and Engineering Undergraduate Scholarship, Fall 2016
Korea Student Aid Foundation
Received full-scholarship for outstanding academic performance in science and engineering

TEACHING
EXPERIENCE

- Teaching Assistant**, Virginia Tech
- Principles Computer Security (CS 4264) Spring 2025
 - Data Structures and Algorithms (CS 3114/5040) Fall 2024
- Teaching Assistant**, Ewha Womans University
- Computer Architecture (20493-02) Fall 2020
 - Covered MIPS assembly language, pipelined CPU, caching, memory systems
 - Led one session for over 60 students, and held weekly office hours
 - Data Structure (20481-01) Fall 2019, 2020
 - Assisted over 90 students with data structure and algorithms with C/C++, Java, Python programming language, marked assignments, and proctored exam
 - Computer Programming and Lab (36339-05) Spring 2020
 - Covered C programming language, RobotC
 - Ran laboratories for 35 students twice a week, had office hours every week, answered questions in online, marked assignments and exam, and proctored exam
 - Computer Algorithm (20497-01) Spring 2019, 2020
 - Discrete Mathematics (20479-02) Spring 2019
- Tutoring**, Ewha Womans University
- One Stop Tutor Fall 2019
 - Tutored programming language for undergraduate students, and held weekly office hours

TECHNICAL
SKILLS

Advanced C/C++, Python, MATLAB, JAVA, L^AT_EX
Moderate MIPS Assembly Language, Android App Programming
Novice JavaScript, HTML, R

RELEVANT
COURSES

Networks Edge Computing, Cloud Computing, Computer Networks
Security Mobile and IoT Security, Cyber Forensics, Software Security
Systems Computer Architecture, Embedded Systems, Operating Systems
Machine Learning Machine Learning, Natural Language Processing, Reinforcement Learning